Satyendra Raj Singh

satendrarajsingh05@gmail.com github.com/srs3161 Phone: +1 (337) 281-9690 LinkedIn

Education

Bachelor of Science, Computer Science 09/2023 – Present
Louisiana Tech University Ruston, LA
Awards: Cyber Technology Practice Program (CTPP) Fellowship
Relevant Coursework: Advanced Data Structures, Computer Architecture, Artificial Intelligence, Systems Programming, Network Security

Experience

IT Help Specialist Ruston, LA
Louisiana Tech University 09/2025 – Present

- Provide front-line technical support to students, faculty, and staff by diagnosing and resolving internet connectivity and computer hardware issues.
- Resolve 10+ weekly IT requests including software installation, hardware troubleshooting, and network connectivity, ensuring minimal downtime.
- Log and track support requests through an IT service ticketing system to maintain accurate records and timely resolution.

Summer Intern Lafayette, LA
Informatics Research Institute, University of Louisiana at Lafayette 06/2024 – 08/2024

- Collaborated on security protocol designs for EV charging stations, contributing to cybersecurity research on connected vehicle infrastructure.
- Worked on hands-on projects focused on cybersecurity solutions for connected vehicles and charging stations.
- Gained practical experience in securing industrial control systems and other emerging technologies.

Projects

TinyML for Person Detection on Microcontrollers Research Project

- Built and evaluated a TinyML person-detection system on **Arduino Nano 33 BLE Sense** using MobileNetV2, pruning, and quantization.
- Compared edge-device inference with GPU-based inference using accuracy, latency, and power metrics.

MNIST Data Classifier Personal Project

- Implemented a feedforward neural network from scratch in **JAVA** for MNIST handwritten digits classification.
- Built model save/load functionality to persist trained weights and biases for later reuse.
- Achieved 96% accuracy on the MNIST test split after multiple training epochs.

Machine Learning-Based Heart Disease Risk Predictor Personal Project

- Built an AI-powered heart disease risk prediction web application that estimates patient risk from clinical inputs in real time.
- Designed a user-friendly interface to simplify data entry and present predictions clearly for non-technical users.

Detection and Tracking of Fast-Moving Tiny Objects Research Project

- Contributed to a research publication manuscript in progress on detecting and tracking fast-moving tiny objects in racquetball video sequences.
- Collected and manually annotated the dataset, and wrote preprocessing scripts to support training and evaluation workflows.
- Evaluated object detection performance using YOLOv5, YOLOv8, and Faster R-CNN to compare robustness on a challenging tiny-object dataset.

Skills

Programming: C, Python, Java, JavaScript, CSS
Frameworks/Tools: PyGame, Tkinter, Flask, PyTorch, OpenCV
Hardware & Other: Arduino, Raspberry Pi, ESP-32, IT Support, Cybersecurity Fundamentals

Certificates

Introduction to Artificial Intelligence with Python – HarvardX / edX
Intro to Java and Object-Oriented Programming – Coursera